# Letter of Motivation

Dr.-Ing. Barış İrhan
Pfaffenwaldring 54
70569 Stuttgart Germany
barisirhan@gmail.com
+49-176-74519826

25.10.2019

To whom it may concern,

I have been graduated from Middle East Technical University (METU), Ankara Turkey, Department of Civil Engineering at 1999. I was a High Honor student with overall GPA 3.57/4.0. I was very enthusiastic and successful about courses on mechanics and numerical methods. During the third-year undergraduate course **CE384-Structural Analysis**, I have been first introduced to finite element method. As part of the course we have developed a finite element program for the analysis of 2D frame structures using Mathcad. In a follow up course **CE483-Advanced Structural Analysis**, we have further improved this program for the analysis of 3D frame structures with options like moment releases, rigid end zones and rigid diaphragms all within the context of small deformations and elastic material behavior. I have taken course **ES504-Numerical Solution of Partial Differential Equations** although it was a graduate course offered for those pursuing their master studies. After I have graduated, I have been employed as a teaching assistant at Structural Mechanics Division of the same department for about 2 years. I was also accepted to Structural Mechanics Division to pursue my master study. In the master I have worked on a topic related with soil structure interaction. Among 7 compulsory courses, I have taken 4 of them from Department of Engineering Science offering courses on theoretical and computational mechanics, namely ES503-Finite Element Method, ES525-Theory of Continuous Media-I, ES526-Theory of Continuous Media-II, ES538-Soil Structure Interaction. After successfully defending my thesis I have been awarded MS degree at 2002.

At 2001 I have been accepted to COMMAS (Computational Mechanics of Materials and Structures) master program organized by Stuttgart University. I was the best student of my class. During the program I have been introduced to Matlab. During my master study I have developed my own finite element program for 1D and 2D continua using Matlab. I have further improved this program later on and used it extensively in my research. In time, it has become an interactive command base program together with some very useful visualization tools. I have done my master thesis at Institute of Applied Mechanics (Chair-I) under the supervision of Prof. Christian Miehe and Dr. Ercan Gürses. The title of my thesis was **"Comparison of alternative kinematical approaches to finite element formulations with embedded strong discontinuities"**. In this study I have implemented 2D finite element formulations for elements cut by strong discontinuities, like cracks, based on elemental and nodal enrichments. I have developed a really original rederivation by myself to motivate locking free finite element formulation based on elemental enrichment, namely statically and kinematically optimal nonsymmetric (SKON) formulation. I have also implemented a damage type traction-separation constitutive law for tractions transmitted through cohesive macroscopic cracks based on internal variable formulation and data structure and algorithms for the global tracking of evolving strong discontinuities. All of the above mentioned implementations have been successfully realized in CMP, an extension of FEAP program, developed and maintained in the institute. After successfully defending my thesis I have been awarded MSc degree at 2003.

After I graduated from COMMAS program I have been offered a research assistant position by Prof. Wolfgang A. Wall at Institute for Computational Mechanics Technical University of Munich to pursue my PhD study. There I have developed coupled Levelset-XFEM formulation for the simulation of

two-phase fluid flows. In addition, I have conducted the diploma thesis of Dr. Peter Gamnitzer, in collaboration with Prof. Wolfgang A. Wall, on overlapping domain decomposition methods for flow simulations. All of the above mentioned formulations have been successfully implemented into finite element program CARAT developed and maintained by the institute, at that time together with Institute for Structural Mechanics University of Stuttgart. The program was written in C programming language. During all of my scientific activities I have made use of and further extended finite element program that I have developed using Matlab as much as possible. Actually, at that time I have decided to name program as MCARAT and offered Prof. Wolfgang A. Wall to maintain this program as an additional tool to be used in research. He did not want to go in that direction by mentioning that it is not feasible to maintain two programs at the same time.

At 2005 I have joined to Institute of Construction Materials University of Stuttgart to pursue my PhD study. There I have worked on finite element formulations for contact and mesh adaptivity. After some trials with MASA program, written in Fortran and maintained by Prof. Josko Ozbolt, I have decided to develop a separate finite element program in Fortran 90. For this purpose first I have developed a finite element program for the implicit simulation of large-deformation frictional and frictionless contact based on Penalty and Augmented-Lagrangian formulations together with adaptive-mesh-refinement and adaptive-remeshing algorithms in 2D and 3D using Matlab. Then I have transferred 3D part of this program together with contact algorithms and remeshing into Fortran. For pre- and post- processing tasks and to perform remeshing, I have developed an interface to Femap program by using api interface provided based on Femap BASIC scripting language. Before I left I have submitted this program to Dr. Vanja Travas who has later been awarded a PhD degree based on a very similar topic.

From 2008 to 2018, I have worked at a software company, PROYA Ankara Turkey as a software engineer. I was the member of a software team gathered to develop COMOSYS (Construction Modelling Systems) program, an object-oriented software for the analysis, design and detailing of commercial and industrial buildings. In this program I was mainly responsible for the development of 3D rebar modelling module. In this company I have learned object oriented programming using C++ programming language, GUI development, basic OpenGL programming and employing Winapi functions to interact with windows operating system.

At 2011 I have been enrolled as a PhD student at Institute of Construction Materials University of Stuttgart under the supervision of Prof. Josko Ozbolt. After successfully defending my thesis I have been awarded Dr.-Ing. degree at 2014. During the period of my study I was in Turkey. I have come to Stuttgart periodically to discuss the current progress. The title of my thesis was **"High velocity impact and fragmentation of concrete: numerical simulation"**. I have taken back the program that I have given to Dr. Vanja Travas which has been further improved by him with explicit time integration, explicit predictor-corrector type Lagrange multiplier contact formulation and updated lagrangian finite element formulation. I have restructured and further improved those algorithms for the simulation of high velocity contact impact events. **Our simulations on perforation of circular plain-concrete slabs with a projectile was the only one in the literature which was able to show perforation limit documented in the experiments**. Within the context of high velocity contact-impact simulations, main ingredients of the finite element program are

- updated Lagrangian finite element formulation,
- explicit time integration,
- predictor-corrector type explicit/implicit contact algorithm,
- element deletion(erosion) algorithm,
- rate and temperature dependent microplane material model at large strains.

During my PhD study, I have developed a finite element analysis program, CIF (Contact Impact and Fragmentation), for 3D high velocity contact impact events. For this purpose, I have developed a program in the form of DLL to perform finite element computations using Intel Fortran. For pre and

post processing tasks I have developed a GUI using Borland C++. I have also developed another program, CIF-JOB, basically to organize simulation projects and perform parametric studies using Borland C++.

During the last year, 08.2018-07.2019, I have worked at Stuttgart University Institute of Construction Materials as a research associate. It was a good opportunity for me to combine my past experiences from computational mechanics and programming in an optimal way. During this period, I have developed a really general-purpose finite element program for the simulation of different physical phenomena. Until I left, I have implemented simulation modules for heat transfer, solid mechanics, explicit dynamics and coupled thermo-mechanics. The program is highly modular and all modules,

- simulations,
- element formulations,
- material models,
- special algorithms (plugins), like contact, element erosion, explicit time integration,
- finite element utilities, like boundary conditions, time curves, initial conditions,

have been implemented in the form of DLLs. The interaction between these different modules has been achieved by client-server type message passing interface. The above-mentioned structure has been constructed for both processor developed in Intel Fortran and pre-post processor developed in Borland C++.

I have a very sound understanding of computational mechanics from both theoretical and implementation point of views. I have been always in touch with either computational mechanics or programming during my academic and professional career. I have a dedicated and goal-oriented attitude. Based on my past experiences, as a student and as a professional, I am very competitive in all aspects related with computational mechanics and programming.

Sincerely yours,

Baris Irhan.